

HRAF DATA FLOW

Simulator (Physics Engine)

Inputs: Joint Commands

- Type: Trajectory Arrays (Radians/Torque)
- Source: IK Execution Layer

Outputs: Camera Frame Feed

- Type: RGB Image Matrix
- Destination: VLM Cognitive Engine

Error States: SimulationFault

VLM Cognitive Engine (AI Brain)

Inputs: Camera Frame Feed

- Type: RGB Image Matrix
- Source: Simulator

Outputs: Target Destination

- Type: Raw Coordinates (JSON Matrix)
- Destination: HARM Module

Error States: JSONFormattingError, TargetHallucination

HARM Module (The Safety Shield)

Inputs: Target Destination

- Type: Raw Coordinates (JSON Matrix)
- Source: VLM Cognitive Engine

Outputs: Verified Safe Target

- Type: Validated Cartesian Coordinates
- Destination: IK Execution Layer

Error States: ValidationError, WorkspaceBoundViolation, LimitExceededError

IK Execution Layer (Mechanical Muscle)

Inputs: Verified Safe Target

- Type: Validated Cartesian Coordinates
- Source: HARM Module

Outputs: Joint Commands

- Type: Trajectory Arrays (Radians/Torque)
- Destination: Simulator

Error States: KinematicSingularity, SelfCollisionFault